METRIC

MIL-L-82893A(OS)
5 August 1994
SUPERSEDING
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MILITARY SPECIFICATION

LEAD FOIL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for lead foil used for chemical purposes in propelling charges for Navy guns (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1169	Packaging, Packing, and Marking for Shipment of Inert Ammunition Components
DOD-STD-2101	Classification of Characteristics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by sending a letter.

AMSC N/A FSC 9535

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* **2.2 Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 37	Chemical Analysis of Pig Lead
E 117	Spectrographic Analysis of Pig Lead by the Point-to-Plane Technique

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials (M101). The raw material shall be such as to produce lead foil meeting the requirements of this specification. The lead foil shall be either primary or secondary lead produced from ore or other materials by a conventional process of reduction and refining or other processes which equally preclude the presence of unusual deleterious elements.

3.2 Chemical composition (M102). The lead foil shall conform to the chemical composition requirements shown in Table I (see 4.3.1).

3.3 Dimensions (M103). The lead foil shall be 0.050 $\pm$ 0.008 mm in thickness. Width shall be 51 cm (20.0 inches) or as specified in the contract or purchase order (see 6.2). Rolls (see 3.4) shall not vary from the ordered width by more than $\pm$ 6.4 mm (see 4.3.2).

3.4 Form. Unless otherwise specified in the contract or purchase order (see 6.2), the lead foil shall be supplied in rolls. The weight of each roll shall not exceed 36 kg (see 4.3.2).

3.5 Workmanship (M104). The lead foil shall be free from dirt, grit, and any other foreign material (see 4.3.2).

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TABLE I. Chemical composition.

Element	Maximum percent
Lead	99.70[1]
Tin	0.1
Antimony	0.07
Arsenic	0.002
Iron	0.002
Bismuth	0.025
Zinc	0.001
Copper	0.04 - 0.08
Silver	0.02

[1] Minimum by difference.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Classification of characteristics. The requirements verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements for critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Requirements not annotated with a classification code are classified as minor.

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4.2 Quality conformance inspection.

4.2.1 Inspection lot. For inspection purposes, a lot shall consist of all lead foil produced by one manufacturer and offered for inspection at one time but shall not exceed 2270 kg. When specified in the contract or purchase order, a lot for chemical analysis shall also be limited to the lead foil produced by one manufacturing or smelting heat. For this case, a lot that cannot be identified by heat shall be cause for rejection.

4.2.2 Sampling for chemical analysis. Two samples for chemical analysis shall be selected from each lot. Each sample shall consist of not less than 600 grams (g) and shall be free from oil, iron, dirt, grit, or other foreign matter. The samples shall be separately packed, labeled, and forwarded to a laboratory for analysis. When a spectrographic analysis is to be made, the quantity and form of the samples shall be as required by the laboratory at which the analysis is to be performed.

4.2.3 Rejections. If a test specimen fails to meet any of the requirements of this specification, it shall be cause for rejection of the lot represented by the specimen. If any lead foil fails to meet the visual or dimensional examination (see 4.3.2), the lot shall be rejected.

* **4.2.4 Retests.** Retests shall be permitted when approved by the responsible Government Engineering Activity.

4.3 Methods of inspection.

* **4.3.1 Chemical analysis.** A chemical analysis shall be made on each sample selected in accordance with 4.2.2 by chemical or spectrographic methods of ASTM E 37 or ASTM E 117 to determine compliance with the requirements of 3.2. In case of a dispute, chemical analysis by chemical methods shall be the basis for acceptance of chemical composition and spectrographic analysis shall be the basis for acceptance of impurity elements of less than 0.01 percent. The contractor shall furnish an analysis (see 6.3) of each melt showing the percentage of each of the elements specified in 3.2. In those instances where the amount of an element allowed by this specification is greater than the concentration range given in either ASTM E 117 (antimony, tin, and iron) or ASTM E 37 (tin and antimony) standard accepted laboratory practices that are suitable for determining the amount of the element shall be employed.

4.3.2 Visual and dimensional inspection. All material in the lot shall be visually and dimensionally examined to verify compliance with the requirements of this specification for form (3.4) and workmanship (3.5). A sufficient number of measurements shall be taken over each roll to ensure that the dimensional tolerances of 3.3 are met.

4.3.4 Inspection of packaging. The packaging, packing, and marking shall be visually examined for conformance to section 5.

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5. PACKAGING

5.1 Packaging. The lead foil shall be packaged in rolls (see 3.4 and 6.2).

* **5.2 Packing.** Packing shall be level A, B, or C, as specified in the contract of purchase order, and shall be in compliance with MIL-STD-1169.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), all markings for shipment shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The lead foil specified herein is intended for use as a gun barrel decoppering agent.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Dimensions if other than as specified (see 3.3).

d. Form, if other than rolls (see 3.4).

e. When lot is to be limited to one heat (see 4.2.1).

f. Analysis report if other than as specified (see 4.3.1).

g. Applicable level of packing (see 5.2).

h. Special marking, if required (see 5.3).

* i. That no ozone depleting substances (ODS) are used in the manufacture of this item.

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6.3 Consideration of data requirements. The following data requirement should be considered when this specification is applied on a contract. The applicable Data Item Description (DID) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID is tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
4.3.1	UDI-T-23191B	Certification data for material

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

* **6.4 Changes from previous issue.** The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.5 Subject term (key word) listing.

Chemical analysis
Decoppering agent
Spectrographic testing

Preparing Activity:
Navy - OS
(Project 9535-N055)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-L-82893A(OS)	2. DOCUMENT DATE (YYMMDD) 940805

3. DOCUMENT TITLE
LEAD FOIL

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER	b. TELEPHONE (*Include Area Code*) (1) Commercial: 301-743-4358/4510 (2) AUTOVON: 354-4358/4510
c. ADDRESS (*Include Zip Code*) CODE 8420 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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